

|                                                 |                                     |                                          |                                                 |                                      |                                      |                                          |
|-------------------------------------------------|-------------------------------------|------------------------------------------|-------------------------------------------------|--------------------------------------|--------------------------------------|------------------------------------------|
|                                                 | $\mathcal{K}_{0^+}^{\#1}$           | $\mathcal{K}_{0^+}^{\#1\alpha\beta\chi}$ |                                                 |                                      |                                      |                                          |
| $\mathcal{K}_{0^+}^{\#1\dagger}$                | $-\frac{3\Upsilon_1 k^2}{2}$        | 0                                        |                                                 |                                      |                                      |                                          |
| $\mathcal{K}_{0^+}^{\#1\dagger\alpha\beta\chi}$ | 0                                   | 0                                        | $\mathcal{K}_{1^+}^{\#1\alpha\beta}$            | $\mathcal{K}_{1^+}^{\#2\alpha\beta}$ | $\mathcal{K}_{1^+}^{\#1\alpha}$      | $\mathcal{K}_{1^+}^{\#2\alpha}$          |
| $\mathcal{K}_{1^+}^{\#1\dagger\alpha\beta}$     | $-k^2 \frac{(4)}{\kappa_3}$         | $\frac{k^2 (4)}{2\sqrt{2}\kappa_4}$      | 0                                               | 0                                    |                                      |                                          |
| $\mathcal{K}_{1^+}^{\#2\dagger\alpha\beta}$     | $\frac{k^2 (4)}{2\sqrt{2}\kappa_4}$ | $-\frac{\Upsilon_2 k^2}{2}$              | 0                                               | 0                                    |                                      |                                          |
| $\mathcal{K}_{1^+}^{\#1\dagger\alpha}$          | 0                                   | 0                                        | $-k^2 \frac{(4)}{\kappa_2}$                     | $-\sqrt{2} k^2 \frac{(4)}{\kappa_2}$ |                                      |                                          |
| $\mathcal{K}_{1^+}^{\#2\dagger\alpha}$          | 0                                   | 0                                        | $-\sqrt{2} k^2 \frac{(4)}{\kappa_2}$            | $-2 k^2 \frac{(4)}{\kappa_2}$        | $\mathcal{K}_{2^+}^{\#1\alpha\beta}$ | $\mathcal{K}_{2^+}^{\#1\alpha\beta\chi}$ |
|                                                 |                                     |                                          | $\mathcal{K}_{2^+}^{\#1\dagger\alpha\beta}$     | 0                                    | 0                                    |                                          |
|                                                 |                                     |                                          | $\mathcal{K}_{2^+}^{\#1\dagger\alpha\beta\chi}$ | 0                                    | 0                                    |                                          |

|                                                 |                                                     |                                                     |                                                 |                                               |                                      |                                          |
|-------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------|-------------------------------------------------|-----------------------------------------------|--------------------------------------|------------------------------------------|
|                                                 | $\mathcal{J}_{0^+}^{\#1}$                           | $\mathcal{J}_{0^+}^{\#1\alpha\beta\chi}$            |                                                 |                                               |                                      |                                          |
| $\mathcal{J}_{0^+}^{\#1\dagger}$                | $\frac{1}{\text{Det}(0^+)}$                         | 0                                                   |                                                 |                                               |                                      |                                          |
| $\mathcal{J}_{0^+}^{\#1\dagger\alpha\beta\chi}$ | 0                                                   | 0                                                   | $\mathcal{J}_{1^+}^{\#1\alpha\beta}$            | $\mathcal{J}_{1^+}^{\#2\alpha\beta}$          | $\mathcal{J}_{1^+}^{\#1\alpha}$      | $\mathcal{J}_{1^+}^{\#2\alpha}$          |
| $\mathcal{J}_{1^+}^{\#1\dagger\alpha\beta}$     | $-\frac{\Upsilon_2 k^2}{2\text{Det}(1^+)}$          | $-\frac{k^2 (4)}{2\sqrt{2}\text{Det}(1^+)\kappa_4}$ | 0                                               | 0                                             |                                      |                                          |
| $\mathcal{J}_{1^+}^{\#2\dagger\alpha\beta}$     | $-\frac{k^2 (4)}{2\sqrt{2}\text{Det}(1^+)\kappa_4}$ | $-\frac{k^2 (4)}{\text{Det}(1^+)\kappa_3}$          | 0                                               | 0                                             |                                      |                                          |
| $\mathcal{J}_{1^+}^{\#1\dagger\alpha}$          | 0                                                   | 0                                                   | $-\frac{1}{9k^2 \frac{(4)}{\kappa_2}}$          | $-\frac{\sqrt{2}}{9k^2 \frac{(4)}{\kappa_2}}$ |                                      |                                          |
| $\mathcal{J}_{1^+}^{\#2\dagger\alpha}$          | 0                                                   | 0                                                   | $-\frac{\sqrt{2}}{9k^2 \frac{(4)}{\kappa_2}}$   | $-\frac{2}{9k^2 \frac{(4)}{\kappa_2}}$        | $\mathcal{J}_{2^+}^{\#1\alpha\beta}$ | $\mathcal{J}_{2^+}^{\#1\alpha\beta\chi}$ |
|                                                 |                                                     |                                                     | $\mathcal{J}_{2^+}^{\#1\dagger\alpha\beta}$     | 0                                             | 0                                    |                                          |
|                                                 |                                                     |                                                     | $\mathcal{J}_{2^+}^{\#1\dagger\alpha\beta\chi}$ | 0                                             | 0                                    |                                          |

#### Abbreviations used in matrices

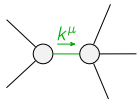
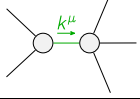
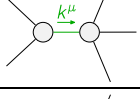
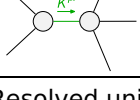
$$\Upsilon_1 == \frac{(4)}{\kappa_1} + \frac{(4)}{\kappa_2} \text{ \&\& } \Upsilon_2 == 9 \frac{(4)}{\kappa_2} - \frac{(4)}{\kappa_3} - \frac{(4)}{\kappa_4} \text{ \&\& }$$

$$\text{Det}(0^+) == -\frac{3}{2} k^2 \left( \frac{(4)}{\kappa_1} + \frac{(4)}{\kappa_2} \right) \text{ \&\& } \text{Det}(1^+) == \frac{1}{8} k^4 (36 \frac{(4)}{\kappa_2} \frac{(4)}{\kappa_3} - (2 \frac{(4)}{\kappa_3} + \frac{(4)}{\kappa_4})^2)$$

#### Lagrangian

$$\frac{(4)}{\kappa_1} \partial_\beta \mathcal{K}_{\chi\delta}^\delta \partial^\chi \mathcal{K}_\alpha^{\alpha\beta} + \frac{(4)}{\kappa_2} \partial_\chi \mathcal{K}_{\beta\delta}^\delta \partial^\chi \mathcal{K}_\alpha^{\alpha\beta} + \frac{(4)}{\kappa_3} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}^\delta + \frac{(4)}{\kappa_4} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta - \frac{(4)}{\kappa_3} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta -$$

$$6 \frac{(4)}{\kappa_2} \partial^\chi \mathcal{K}_\alpha^{\alpha\beta} \partial_\delta \mathcal{K}_{\chi\beta}^\delta - \frac{9}{2} \frac{(4)}{\kappa_2} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta + \frac{1}{2} \frac{(4)}{\kappa_3} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta + \frac{1}{2} \frac{(4)}{\kappa_4} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta$$

|                                                                                    |                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Added source term(s):                                                              | $\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$ |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                         |
| Source constraint(s)                                                               | # constraint(s)                                               | Covariant form                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                         |
| $\mathcal{J}_{1^+}^{\#1\alpha} == \mathcal{J}_{1^+}^{\#2\alpha}$                   | 3                                                             | $\partial_\chi \partial^\alpha \mathcal{J}_{\beta\chi}^{\beta\chi} + \partial_\chi \partial^\chi \mathcal{J}^{\beta\alpha}_{\beta} == 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                         |
| $\mathcal{J}_{2^+}^{\#1\alpha\beta\chi} == 0$                                      | 5                                                             | $3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta}_{\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} +$ $4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\alpha \mathcal{J}^{\delta}_{\delta}{}^\epsilon + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\beta\epsilon} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{J}^{\delta\alpha}_{\delta} ==$ $3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha}_{\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} +$ $2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\beta \mathcal{J}^{\delta}_{\delta}{}^\epsilon + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\alpha\epsilon} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{J}^{\delta\beta}_{\delta}$ |                                                                                                                                                                                                                                                                                                                                                                                                         |
| $\mathcal{J}_{2^+}^{\#1\alpha\beta} == 0$                                          | 5                                                             | $3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \eta^{\alpha\beta} \partial_\epsilon \partial^\epsilon \partial_\delta \mathcal{J}^{\chi}_{\chi}{}^\delta == 2 \partial_\delta \partial^\beta \partial^\alpha \mathcal{J}^{\chi}_{\chi}{}^\delta + 3 (\partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                         |
| Total # constraint(s):                                                             | 13                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                         |
| Resolved pole(s)                                                                   | # polarization(s)                                             | Square mass                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Residue                                                                                                                                                                                                                                                                                                                                                                                                 |
|  | 1                                                             | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $-\frac{216 \frac{(4)}{\kappa_2}{}^2 - 24 \frac{(4)}{\kappa_2} (2 \frac{(4)}{\kappa_3} + \frac{(4)}{\kappa_4}) + (2 \frac{(4)}{\kappa_3} + \frac{(4)}{\kappa_4})^2}{\frac{(4)}{\kappa_2} (36 \frac{(4)}{\kappa_2} \frac{(4)}{\kappa_3} - (2 \frac{(4)}{\kappa_3} + \frac{(4)}{\kappa_4})^2)}$                                                                                                           |
|  | 1                                                             | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $-\frac{48 \frac{(4)}{\kappa_2}{}^2 - 8 \frac{(4)}{\kappa_2} (2 \frac{(4)}{\kappa_3} + \frac{(4)}{\kappa_4}) + (2 \frac{(4)}{\kappa_3} + \frac{(4)}{\kappa_4})^2}{\frac{(4)}{\kappa_2} (36 \frac{(4)}{\kappa_2} \frac{(4)}{\kappa_3} - (2 \frac{(4)}{\kappa_3} + \frac{(4)}{\kappa_4})^2)}$                                                                                                             |
|  | 1                                                             | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $-\frac{\frac{(4)}{\kappa_2} (2 \frac{(4)}{\kappa_3} + \frac{(4)}{\kappa_4}) + \sqrt{\frac{(4)}{\kappa_2}{}^2 (81 \frac{(4)}{\kappa_2}{}^2 - 18 \frac{(4)}{\kappa_2} (2 \frac{(4)}{\kappa_3} + \frac{(4)}{\kappa_4}) + 4 (2 \frac{(4)}{\kappa_3} + \frac{(4)}{\kappa_4})^2)}}{\frac{(4)}{\kappa_2} (36 \frac{(4)}{\kappa_2} \frac{(4)}{\kappa_3} - (2 \frac{(4)}{\kappa_3} + \frac{(4)}{\kappa_4})^2)}$ |
|  | 1                                                             | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $-\frac{\frac{(4)}{\kappa_2} (2 \frac{(4)}{\kappa_3} + \frac{(4)}{\kappa_4}) + \sqrt{\frac{(4)}{\kappa_2}{}^2 (81 \frac{(4)}{\kappa_2}{}^2 - 18 \frac{(4)}{\kappa_2} (2 \frac{(4)}{\kappa_3} + \frac{(4)}{\kappa_4}) + 4 (2 \frac{(4)}{\kappa_3} + \frac{(4)}{\kappa_4})^2)}}{\frac{(4)}{\kappa_2} (36 \frac{(4)}{\kappa_2} \frac{(4)}{\kappa_3} - (2 \frac{(4)}{\kappa_3} + \frac{(4)}{\kappa_4})^2)}$ |
| Resolved unitarity condition(s):                                                   | (Demonstrably impossible)                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                         |